Differential Equations — Reference

how to read this note

Every method is a chain of steps. Each step is a template with empty colored slots.

Do	How
1. find the pieces	Each lesson opens with a PIECES table. It says where each piece is in your equation, and its color.
2. fill the slots	A piece goes in the slot of the same color: p goes in $\boxed{p}$.
3. one box at a time	Each box is one move. Do not merge two boxes.

Colors in lessons 1, 3, 5 and Bernoulli: $p(x)$ $q(x)$ μ (mu) x_0, y_0 $a(x)$ or n. Lessons 2, 4, 10–15 and the homogeneous method set their own colors in their PIECES tables.

the master idea

Rearrange until each side is integrable. Second order: guess the shape of y.

Method	Rearrange (or guess) until
linear (L3)	the left side is one derivative, $(\mu y)'$
separable (L4)	all y is with dy, all x is with dx
Bernoulli (L6)	it is linear in $v = y^{1-n}$. Then L3
homogeneous (L6)	it is separable in $v = y/x$. Then L4
exact (L7)	the left side is one differential, dF
autonomous (L8)	no solving: read the phase line
const. coefficients (L10)	guess e^{rx}: it is a quadratic in r
Euler (L11)	guess x^r: it is a quadratic in r
nonhomogeneous (L12)	it splits: $y = y_c + y_p$
undet. coefficients (L13)	a guess shaped like f fits: match the numbers
reduction of order (L14)	$y = uy_1$ drops to first order in u'
variation of param. (L15)	$y_p = u_1y_1 + u_2y_2$: two integrals

which method? — the sorter

Test in this order. Stop at the first yes.

If the equation	Use	
fits $y' + p(x)\,y = q(x)$ after you divide	linear, seven lines	(L3)
factors as $y' = g(x)\,h(y)$	separable	(L4)
fits $y' + py = qy^n$, $n \neq 0, 1$	Bernoulli, $v = y^{1-n}$	(L6)
has a right side in $\frac{y}{x}$ only	homogeneous, $v = \frac{y}{x}$	(L6)
is $M\,dx + N\,dy = 0$ with $M_y = N_x$	exact, build F	(L7)
has $M_y \neq N_x$	integrating factor $\mu(x)$ or $\mu(y)$	(L7)

Trap: Some equations fit two rows. $y' + 3y = 6$ is linear and separable. Either method works.
Trap: Bernoulli with $n = 0$ or $n = 1$ is already linear. Do not substitute.

which method? — second order

Move all y-terms to the left. Then test.

If the equation	Use	
is $ay'' + by' + cy = 0$, a, b, c numbers	characteristic equation	(L10)
is $ax^2y'' + bxy' + cy = 0$	indicial equation	(L11)
has a right side $f(x) \neq 0$	y_c from L10 or L11, $+\,y_p$	(L12)
numbers, f from x^n, e^{kx}, cos, sin	y_p by undetermined coefficients	(L13)
any other f, or not numbers	y_p by variation of parameters	(L15)
has no recipe, but one y_1 is given	y_2 by reduction of order	(L14)

Trap: $y'' + xy' + y = 0$ is neither: the coefficients are not numbers, and the powers of x do not match the orders.

which way? — differentiate or integrate

Task	Direction
verify a solution	differentiate the given y
check μ	differentiate μ, compare with $p\mu$
uniqueness test	differentiate f in y (x is a constant)
linear line 6, separable step 3	integrate

Trap: To verify, never solve the DE for y'. Get y' from the given y.

which tool? — given → find

Given → find	Do this	
an equation → order	highest derivative	(L1)
an equation → linear or not	the linear checklist	(L1)
an equation → p and q	standard form	(L1)
a y and a DE → solution or not	differentiate y, substitute	(L1)
$y' = f$ and a point → slope	evaluate f there	(L2)
$f \to$ zero-slope set	factor $f = 0$	(L2)
f with no x, y_0 → limit	nearest flat line in the drift	(L2)
linear → general solution	seven lines	(L3)
separable → general solution	separate, integrate, free y	(L4)
general solution, IC → C	plug in x_0, y_0	(L3, L4)
separable → lost solutions	roots of $h(y) = 0$	(L4)
linear IVP → interval	bad x of p and q	(L5)
nonlinear IVP → unique or not	Test 1, Test 2	(L5)
Bernoulli → solution	$v = y^{1-n}$, then L3	(L6)
$F(y/x) \to$ solution	$v = y/x$, then L4	(L6)
$M\,dx + N\,dy \to$ exact or not	$M_y = N_x$	(L7)
exact → solution	build F, answer $F = C$	(L7)
not exact → multiplier	one-letter ratio test	(L7)
$y' = f(y) \to$ equilibria, stability	phase line	(L8)
$y' = f(y)$, $y_0 \to$ limit	follow the arrow	(L8)
words → a DE	the translation table	(L9)
a second reading → k	plug in, take ln	(L9)
tank → $Q(t)$	in − out, linear	(L9)
$ay'' + by' + cy = 0 \to$ solution	roots of $ar^2 + br + c$, sign of D	(L10)
Euler → general solution	roots of $ar^2 + (b-a)r + c$	(L11)
$y_1, y_2 \to$ fundamental set or not	$W \neq 0$	(L12)
$f \neq 0 \to$ general solution	$y_c + y_p$	(L12)
2nd-order IVP → c_1, c_2; interval	y, y' at x_0; bad x of p, q, f	(L12)
numbers, f in the family → y_p	guess $\times x^s$, match	(L13)
one $y_1 \to y_2$	$y = uy_1$, $w = e^{-\int p}/y_1^2$	(L14)
y_1, y_2, any $f \to y_p$	$u_1' = -\frac{y_2 f}{W}$, $u_2' = \frac{y_1 f}{W}$	(L15)

integrals (add $+C$ once)

Function	Antiderivative
x^n, $n \neq -1$	$\frac{x^{n+1}}{n+1}$
y^{-n}, $n \neq 1$	$\frac{y^{1-n}}{1-n}$ *(only $n = 1$ gives ln)*
$\frac{k}{x}$	$k \ln\lvert x\rvert$ not $\ln \frac{k}{x}$
$\frac{1}{ax+b}$	$\frac{1}{a} \ln\lvert ax+b\rvert$
e^{kx}	$\frac{1}{k} e^{kx}$ divide by k
$\sin(ax)$ / $\cos(ax)$	$-\frac{\cos(ax)}{a}$ / $\frac{\sin(ax)}{a}$
$\tan x$	$\ln\lvert\sec x\rvert$
$\sec^2 x$	$\tan x$
$\sec x$	$\ln\lvert\sec x + \tan x\rvert$ *(L15)*
xe^{kx}	$\frac{1}{k^2}(kx-1)\,e^{kx}$ *(by parts)*
$\frac{1}{1+y^2}$	$\tan^{-1} y$

By parts: $\int u\,dv = uv - \int v\,du$. Pick u = the part that gets simpler when you differentiate (x, x^2, $\ln x$).
Product rule (the heart of L3): $(\mu y)' = \mu' y + \mu y'$.

exponent and log rules

Rule	Use
$e^{k\ln u} = u^k$	simplify μ: $e^{2\ln x} = x^2$, $e^{-\ln x} = \frac{1}{x}$
$\frac{1}{e^{-kx}} = e^{kx}$	line 7: divide by μ
$\frac{1}{\sec x} = \cos x$	line 7 when $\mu = \sec x$
$e^{a+C} = e^C e^a$	L4: rename $e^C \to C$
e^{ab}	does not split: $e^{xy} \neq e^x e^y$
$\ln(e^{\square}) = \square$	ln and e undo each other
$e^{(\alpha+\beta i)x} = e^{\alpha x}(\cos\beta x + i\sin\beta x)$	Euler's formula: L10 complex roots
$x^r = e^{r\ln x}$	L11: $x^{\beta i} = \cos(\beta\ln x) + i\sin(\beta\ln x)$

Lesson 1 — classify, standard form, verify

EXPLANATION

THE IDEA
Label the equation before you solve it: order, linear or not, homogeneous or not. Then put it in standard form, so that you can read p and q.

Pieces

Piece	Where to find it
$a(x)$	the coefficient of y', with all y-terms on the left
$p(x)$	the coefficient of y, after you divide by a. It keeps its sign
$q(x)$	the right side, after you divide by a

Read the pieces

Quick example	Work → answer
$xy'+2y=x^2$	$a=x$: $y'+\frac{2}{x}y=x$
$y'=3y+x$	move: $y'-3y=x$. $p=-3$
$y'+y\sin x=0$	$p=\sin x$, $q=0$: homogeneous
$y'=xy^2$	y^2: not linear. No p, no q

Linear or not?

Still linear	Breaks linearity
x^2y', e^xy, $\sin x$	y^2, yy', $(y')^3$
a missing term (coefficient 0)	$\sin y$, e^y

Trap: p includes its sign. $y'-3y$ has $p=-3$.
Trap: Divide every term by a, the right side too.
Trap: Homogeneous means q is the zero function. Do not plug in $x=0$.
Trap: To verify, never solve the DE for y'. Get y' from the given y.

WORK

Steps — classify

1. ORDER
The highest derivative present. Exponents do not matter: $(y'')^3$ is order 2.

↓

2. LINEAR?
Every $y, y', y'', \ldots$ appears alone, to the first power, with a coefficient in x only. Never x-stuff breaks it.

↓

3. HOMOGENEOUS? (2.1 SENSE)
Move all y-terms to the left. If the right side is the zero function, it is homogeneous.

Steps — standard form

1. MOVE
All y-terms to the left: $\boxed{a}y'+by=g$

↓

2. DIVIDE EVERY TERM BY a
$y'+\boxed{\frac{b}{a}}y=\boxed{\frac{g}{a}}$ *(the coefficient of y' must be 1)*

↓

3. READ
$p=\frac{b}{a}$ $\quad q=\frac{g}{a}$ *(write both down)*

Steps — verify a solution

1. DIFFERENTIATE
Find y' (and y'' if needed) from the given y, with calculus.

↓

2. SUBSTITUTE
Put y, y', … into the left side only.

↓

3. SIMPLIFY AND COMPARE
Simplify the left side. If it equals the right side for every x, y is a solution.

Lesson 2 — direction fields

EXPLANATION

THE IDEA
$y' = f(x,y)$ gives the slope at every point. The field is a picture of those slopes. You never solve anything.

Pieces

Piece	Where to find it
$f(x,y)$	the right side, after you get y' alone
(x_0,y_0)	the point you are asked about

Sign of the slope

Slope	The solution
$y' > 0$	rises as x increases
$y' < 0$	falls
$y' = 0$	is flat

Does f have x?

If f	Then
has no x: $y' = f(y)$	every column is the same. Zero-slope lines are horizontal (equilibria). Solutions approach them and never cross
has x	the zero-slope set is a curve, not a wall. No equilibria

Trap: Never divide $f = 0$ by y. You lose the root $y = 0$.
Trap: The limit only works when f has no x.

WORK

Steps — slope at a point

1. TEMPLATE
slope $= \boxed{f}\left(\boxed{x_0}, \boxed{y_0}\right)$

↓

2. READ THE SIGN
$+$ rises, $-$ falls, 0 flat.

Steps — zero-slope set

1. SET
$\boxed{f(x,y)} = 0$

↓

2. FACTOR
Pull out every common factor. Do not divide.

↓

3. EACH FACTOR $= 0$
Each factor gives one line or curve of the set.

Steps — limit as $x \to \infty$ (f has no x)

1. FLAT LINES
Solve $\boxed{f(y)} = 0$. List the roots.

↓

2. DRIFT
Sign of $\boxed{f}\left(\boxed{y_0}\right)$: $+$ moves up, $-$ moves down, 0 stays.

↓

3. LIMIT
The nearest flat line in the direction of drift. If there is none in that direction: $\pm\infty$.

Lesson 3 — first order linear (seven lines)

EXPLANATION

THE IDEA

Multiply by μ so that the left side becomes one product-rule derivative, $(\mu y)'$. Then one integral finishes the job.

Pieces

Piece	Where to find it
p	the coefficient of y in standard form, sign included
q	the right side in standard form
μ (mu)	you make it: $e^{\int p\,dx}$. No $+C$
x_0, y_0	the initial condition. Use it only at the end

Line 2 — common μ

If p is	then μ is
k (a constant)	e^{kx}
$\frac{k}{x}$	$e^{k\ln x} = x^k$
$-\frac{1}{x}$	$e^{-\ln x} = \frac{1}{x}$
$2x$	e^{x^2}
$\tan x$	$e^{\ln\lvert\sec x\rvert} = \sec x$
$-\tan x$	$e^{\ln\lvert\cos x\rvert} = \cos x$

Equivalences

This	is the same as
μ'	$p\mu$, always (that is line 3)
$\mu p\,y$	a line 4 term
μy	the line 6 left side
$\frac{1}{e^{-kx}}$ / $\frac{1}{\sec x}$	e^{kx} / $\cos x$

Trap: $\int \frac{k}{x}\,dx = k\ln x$, not $\ln\frac{k}{x}$.
Trap: $\int e^{kx}dx = \frac{1}{k}e^{kx}$. Divide by k.
Trap: $\int \tan x\,dx = \ln\lvert\sec x\rvert$.
Trap: Line 7: divide C by μ too.

WORK

The seven lines

LINE 1 — STANDARD FORM, READ THE PIECES

$y' + \boxed{p}y = \boxed{q}$ Write $p = \ldots$ and $q = \ldots$

↓

LINE 2 — BUILD μ

$\boxed{\mu} = e^{\int \boxed{p}dx}$ Simplify. $(e^{k\ln u} = u^k)$ No $+C$.

↓

LINE 3 — CHECK μ

Differentiate: $\boxed{\mu}' = \ldots$ Separately: $\boxed{p}\boxed{\mu} = \ldots$
They must be equal. If not, fix line 2.

↓

LINE 4 — MULTIPLY EVERY TERM

$\boxed{\mu}y' + \boxed{\mu}\boxed{p}y = \boxed{\mu}\boxed{q}$
(three multiplications, written out)

↓

LINE 5 — COLLAPSE THE LEFT SIDE

$\left(\boxed{\mu}y\right)' = \boxed{\mu}\boxed{q}$
$((\mu y)' = \mu' y + \mu y' = \mu p\,y + \mu y'$: the whole left side of line 4)

↓

LINE 6 — INTEGRATE BOTH SIDES

$\boxed{\mu}y = \int \boxed{\mu}\boxed{q}dx + C$
(left side is μ times y: not μ', not μp)

↓

LINE 7 — DIVIDE BY μ

$y = \frac{1}{\boxed{\mu}}\left(\int \boxed{\mu}\boxed{q}dx + C\right)$
(divide the whole parenthesis: C becomes C/μ)

↓

IC (IF GIVEN)

Plug in x_0 and y_0. Solve for C. Rewrite the answer with the number.

Lesson 4 — separable

EXPLANATION

THE IDEA

Factor the right side into an x-part times a y-part. Move all y to the dy side and all x to the dx side. Integrate each side.

Pieces

Piece	Where to find it
$g(x)$	the factor with only x (a constant counts)
$h(y)$	the factor with only y
x_0, y_0	the initial condition

Read the pieces

Quick example	Work → answer
$y' = xy$	$g = x$, $h = y$
$y' = \frac{x}{y}$	$g = x$, $h = \frac{1}{y}$
$y' = e^{x+y}$	$= e^x e^y$: $g = e^x$, $h = e^y$
$y' = 2y$	$g = 2$, $h = y$
$y' = x + y$	will not factor: not separable

Step 4 — free y

Wrapped	Undo with	Watch for
$\ln\lvert y\rvert = \ldots$	$e^{(\cdot)}$ of the whole side	$e^{a+C} = e^C e^a$. Rename $e^C \to C$
$y^2 = \ldots$	square root	$y = \pm\sqrt{\ldots}$. Keep $\pm$ unless an IC picks one
$\frac{1}{y} = \ldots$	reciprocal	do it first, before anything else
$\tan^{-1} y = \ldots$	tan	$y = \tan(\ldots + C)$, C stays inside

Trap: $\int y^{-n}\,dy = \frac{y^{1-n}}{1-n}$ for $n \neq 1$. Only $n = 1$ gives ln.
Trap: Keep the dy and the dx when you separate.

WORK

Steps

1. FACTOR

$\frac{dy}{dx} = \boxed{g(x)}\,\boxed{h(y)}$ If it will not factor, it is not separable.

↓

2. SEPARATE

$\frac{dy}{\boxed{h(y)}} = \boxed{g(x)}\,dx$

↓

3. INTEGRATE

$\int \frac{dy}{\boxed{h(y)}} = \int \boxed{g(x)}\,dx + C$ *(one $+C$ only)*

↓

4. FREE y

Apply the inverse of whatever wraps y. Use the table on the left.

↓

5. IC (IF GIVEN)

Plug in x_0, y_0. Solve for C.

↓

6. LOST SOLUTIONS

Solve $\boxed{h(y)} = 0$. Each root $y = k$ is a constant solution. Step 2 divided by it. Mention them if asked for all solutions.

Lesson 5 — existence and uniqueness

THE IDEA
Before you solve, ask: does a solution through the point exist, and is it the only one? Linear and nonlinear equations use different tests.

Linear: the interval

EXPLANATION

Pieces

Piece	Where to find it
p, q	standard form $y' + py = q$
x_0	the IC $y(x_0) = y_0$. y_0 does not matter

Bad x

Undefined	Bad x
division by 0	where the bottom $= 0$
$\ln(\dots)$	where $\dots \le 0$
even root $\sqrt{\dots}$	where $\dots < 0$
$\tan x$	$x = \frac{\pi}{2} + n\pi$

Read the pieces

Quick example	Work → answer
$p = \frac{1}{x-3}$, $x_0 = 1$	bad: 3. $(-\infty, 3)$
$p = \ln x$, $x_0 = 2$	bad: $x \le 0$. $(0, \infty)$

Trap: Divide by $a(x)$ first. Its zeros become bad x.

WORK

1. STANDARD FORM
$y' + \boxed{p}\,y = \boxed{q}$

↓

2. BAD x OF p
List every x where p is undefined.

↓

3. BAD x OF q
List every x where q is undefined.

↓

4. NUMBER LINE
Mark all bad x and x_0.

↓

5. INTERVAL
(nearest bad x below x_0, nearest bad x above x_0). Open. No bad x on a side: $\pm\infty$.

Nonlinear: Test 1 and Test 2

EXPLANATION

Pieces

Piece	Where to find it
$f(x,y)$	the right side of $y' = f(x,y)$
$\frac{\partial f}{\partial y}$	you make it: differentiate f in y, x held constant
(x_0, y_0)	the IC. Both numbers matter

Verdict

Result	Verdict
both pass	a unique solution
Test 1 yes, Test 2 no	exists, uniqueness not guaranteed
Test 1 fails	the theorem says nothing

Read the pieces

Quick example	Work → answer
$f = y^{1/3}$, $y_0 = 0$	$\frac{1}{3y^{2/3}}$ breaks at 0: exists, not guaranteed unique
$f = y^{1/3}$, $y_0 = 1$	both pass: unique

Trap: The tests are at the point. Changing y_0 can change the verdict.
Trap: A point on the edge of the domain fails Test 1: $\sqrt{y}$ at $y_0 = 0$ is not defined below.

WORK

1. NORMAL FORM
$y' = \boxed{f(x,y)}$

↓

2. TEST 1 — EXISTENCE
$\boxed{f}\left(\boxed{x_0}, \boxed{y_0}\right)$ is a real number, and f is defined on an open rectangle around the point.

↓

3. PARTIAL
$\boxed{\frac{\partial f}{\partial y}} = \dots$ *(x is a constant)*

↓

4. TEST 2 — UNIQUENESS
$\boxed{\frac{\partial f}{\partial y}}\left(\boxed{x_0}, \boxed{y_0}\right)$ is a real number.

↓

5. VERDICT
Use the verdict table on the left.

Lesson 6 — transformations

THE IDEA
Substitute a new variable so that the equation becomes a type you can already solve. Bernoulli → linear (L3). Homogeneous → separable (L4).

Bernoulli: $y' + p\,y = q\,y^n$

EXPLANATION

Pieces

Piece	Where to find it
p	the coefficient of y on the left
q	the coefficient of y^n on the right
n	the power on the right. $n \neq 0, 1$
new p, new q	$(1-n)\,p$ and $(1-n)\,q$ (step 5)

Read the pieces

Quick example	Work → answer
$y' + y = xy^3$	1, x, 3: $v = y^{-2}$. new -2, new $-2x$
$y' - \frac{y}{x} = y^2$	$-\frac{1}{x}$, 1, 2: $v = y^{-1}$. new $\frac{1}{x}$, new -1

Trap: Keep the $(1-n)$. It is often negative.
Trap: Divide every term by y^n, the y' term too.
Trap: Forget the $(1-n)$ in step 5 and every later line is wrong.

WORK

1. MATCH, READ THE PIECES
$y' + \boxed{p}\,y = \boxed{q}\,y^{\boxed{n}}$

↓

2. CHOOSE v
$v = y^{1-\boxed{n}} \qquad v' = (1-\boxed{n})\,y^{-\boxed{n}}\,y'$

↓

3. DIVIDE EVERY TERM BY y^n
$y^{-\boxed{n}}y' + \boxed{p}\,y^{1-\boxed{n}} = \boxed{q}$

↓

4. REPLACE
$y^{-n}y' \to \dfrac{v'}{1-n}, \quad y^{1-n} \to v: \qquad \dfrac{v'}{1-\boxed{n}} + \boxed{p}\,v = \boxed{q}$

↓

5. MULTIPLY BY $(1-n)$, READ NEW PIECES
$v' + \underbrace{(1-\boxed{n})\boxed{p}}_{\text{new } p}\,v = \underbrace{(1-\boxed{n})\boxed{q}}_{\text{new } q}$

Step 6 — the seven lines on v

LINE 1 — STANDARD FORM IN v
$v' + \boxed{\text{new } p}\,v = \boxed{\text{new } q}$ *(from step 5)*

↓

LINE 2 — BUILD μ
$\boxed{\mu} = e^{\int \boxed{\text{new } p}\,dx}$ Simplify. No $+C$.

↓

LINE 3 — CHECK μ
Differentiate: $\boxed{\mu}' = \ldots$ Separately: $\boxed{\text{new } p}\,\boxed{\mu} = \ldots$
They must be equal.

↓

LINE 4 — MULTIPLY EVERY TERM
$\boxed{\mu}\,v' + \boxed{\mu}\,\boxed{\text{new } p}\,v = \boxed{\mu}\,\boxed{\text{new } q}$

↓

LINE 5 — COLLAPSE THE LEFT SIDE
$\left(\boxed{\mu}\,v\right)' = \boxed{\mu}\,\boxed{\text{new } q}$

↓

LINE 6 — INTEGRATE BOTH SIDES
$\boxed{\mu}\,v = \int \boxed{\mu}\,\boxed{\text{new } q}\,dx + C$ *(μ times v, not y)*

↓

LINE 7 — DIVIDE BY μ
$v = \dfrac{1}{\boxed{\mu}}\left(\int \boxed{\mu}\,\boxed{\text{new } q}\,dx + C\right)$ *(C becomes C/μ)*

↓

7. BACK-SUBSTITUTE
$y = v^{1/(1-\boxed{n})}$ *(raise the whole line 7 answer, C included)*

↓

8. IC (IF GIVEN)
Plug x_0, y_0 into the y answer. Solve for C.

Homogeneous: $y' = F(y/x)$

EXPLANATION

Pieces

Piece	Where to find it
v	$v = \frac{y}{x}$, so $y = vx$
$F(v)$	the right side, rewritten in $\frac{y}{x}$ only, with $\frac{y}{x} \to v$

Get F: divide the top and the bottom by the highest power of x.

Quick example	Work → answer
$y' = \dfrac{x+y}{x}$	$1 + \frac{y}{x}$: $F = 1 + v$
$y' = \dfrac{x^2+y^2}{xy}$	$\div x^2$: $F = \frac{1+v^2}{v}$
$y' = \dfrac{y}{x} + \dfrac{x}{y}$	$F = v + \frac{1}{v}$

Trap: y' is not v'. $y' = v + xv'$.

Two meanings of "homogeneous"

Where	Means
Lesson 1 (Trench 2.1)	the right side of $y' + py = q$ is the zero function: $q = 0$
Lessons 10–12	the same idea: the right side of $y'' + py' + qy = f$ is $f = 0$
Lesson 6	the right side is a function of $\frac{y}{x}$ only: $y' = F(y/x)$

WORK

1. REWRITE
$y' = \boxed{F}\left(\frac{y}{x}\right)$

↓

2. SUBSTITUTE
$y = \boxed{v}x \qquad y' = \boxed{v} + x\boxed{v}'$ *(product rule)*

↓

3. PLACE AND CANCEL
$\boxed{v} + xv' = \boxed{F(v)} \;\Rightarrow\; xv' = \boxed{F(v)} - \boxed{v}$

↓

4. SEPARATE (LESSON 4)
$\dfrac{dv}{\boxed{F(v)} - \boxed{v}} = \dfrac{dx}{x}$ $(g(x) = \frac{1}{x},\ h(v) = F(v) - v)$

↓

5. INTEGRATE
Both sides. One $+C$. $\int \frac{dx}{x} = \ln|x|$.

↓

6. BACK-SUBSTITUTE
Replace v with $\frac{y}{x}$. Free y if you can.

Lesson 7 — exact equations

THE IDEA
Find one function $F(x,y)$ whose partial derivatives are M and N. Then the whole equation says $dF = 0$, so the answer is $F = C$.

Exact: build F

EXPLANATION

Pieces

Piece	Where to find it
$M(x,y)$	whatever multiplies dx
$N(x,y)$	whatever multiplies dy
$g(y)$	the "constant" of the x-integral: an unknown function of y
$F(x,y)$	you build it: $F_x = M$, $F_y = N$

Read the pieces

Quick example	Work → answer
$(2xy+1)\,dx + (x^2+3y^2)\,dy = 0$	$M = 2xy+1$, $N = x^2+3y^2$
$y' = -\dfrac{x}{y}$	$y\,dy = -x\,dx \to x\,dx + y\,dy = 0$: $M = x$, $N = y$

Equivalences

This	is the same as
M	F_x
N	F_y
the test $M_y = N_x$	mixed partials agree: $F_{xy} = F_{yx}$

Trap: Integrate M in x: y is a constant. The "$+C$" is $g(y)$, not a number.
Trap: In step 5, every x term must cancel. If one is left, a step above is wrong.
Trap: The answer is $F = C$, not $F = 0$. Implicit is fine.

WORK

1. FORM, READ THE PIECES
$\boxed{M}\,dx + \boxed{N}\,dy = 0$

↓

2. TEST
$\dfrac{\partial \boxed{M}}{\partial y} \stackrel{?}{=} \dfrac{\partial \boxed{N}}{\partial x}$ Equal → exact. Not equal → stop, or find μ (below).

↓

3. INTEGRATE M IN x
$\boxed{F} = \displaystyle\int \boxed{M}\,dx + \boxed{g(y)}$ *(y held constant)*

↓

4. DIFFERENTIATE F IN y
$\dfrac{\partial \boxed{F}}{\partial y} = \ldots + \boxed{g'(y)}$

↓

5. SET EQUAL TO N, SOLVE FOR g'
$\ldots + \boxed{g'(y)} = \boxed{N}$ *(all x terms cancel)*

↓

6. INTEGRATE g'
$\boxed{g(y)} = \displaystyle\int \boxed{g'(y)}\,dy$ Put it back into step 3.

↓

7. ANSWER
$\boxed{F(x,y)} = C$ IC given: plug in x_0, y_0 for C.

Not exact: find an integrating factor μ

EXPLANATION

Pieces

Piece	Where to find it
M, N	as above
M_y, N_x	from the failed test
μ	you make it: $\mu(y)$ or $\mu(x)$, one letter only

Definition: μ is an integrating factor if $\mu M\,dx + \mu N\,dy = 0$ is exact.

Which μ?

If this has only one letter	then
$\dfrac{N_x - M_y}{M}$ has only y	$\mu(y) = e^{\int (\text{that})\,dy}$
$\dfrac{M_y - N_x}{N}$ has only x	$\mu(x) = e^{\int (\text{that})\,dx}$

Trap: $\mu(y)$ has no x, so $\frac{\partial}{\partial x}(\mu N) = \mu N_x$. Only the y-side needs the product rule.
Trap: No $+C$ in μ.

WORK

1. MULTIPLY
$\boxed{\mu}\boxed{M}\,dx + \boxed{\mu}\boxed{N}\,dy = 0$

↓

2. REQUIRE EXACT
$\dfrac{\partial(\boxed{\mu}\boxed{M})}{\partial y} = \dfrac{\partial(\boxed{\mu}\boxed{N})}{\partial x}$

↓

3. EXPAND, FOR $\mu = \mu(y)$
$\boxed{\mu}'\boxed{M} + \boxed{\mu}\boxed{M_y} = \boxed{\mu}\boxed{N_x}$ *(product rule on the left only)*

↓

4. SEPARATE (LESSON 4)
$\dfrac{\boxed{\mu}'}{\boxed{\mu}} = \dfrac{\boxed{N_x} - \boxed{M_y}}{\boxed{M}}$ Must have only y.
$\boxed{\mu} = e^{\int (\ldots)\,dy}$

↓

5. NEW PIECES, REDO
new $M = \mu M$, new $N = \mu N$. Do exact steps 2–7.

Lesson 8 — autonomous equations

EXPLANATION

THE IDEA

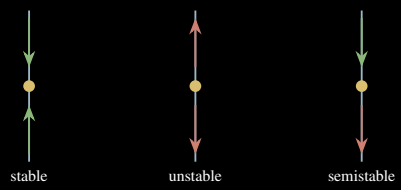
$\frac{dy}{dt} = f(y)$ has no t on the right. This is the Lesson 2 "no x" case. You read the long-term behavior off a phase line, without solving.

Pieces

Piece	Where to find it
$f(y)$	the right side. No t in it
equilibria	the roots of $f(y) = 0$: constant solutions
y_0	the IC $y(0) = y_0$

Classify an equilibrium

Arrows	Type
both point toward it	stable (asymptotically stable, sink)
both point away	unstable (source)
one toward, one away	semistable



Sign → arrow

Sign of f	Arrow
$f > 0$	up (y increases)
$f < 0$	down (y decreases)

Trap: Never divide $f = 0$ by y. You lose the equilibrium $y = 0$.
Trap: Test one y inside each region, not at an equilibrium.
Trap: A solution never crosses an equilibrium.

WORK

1. EQUILIBRIA
Solve $\boxed{f(y)} = 0$. Factor, set each factor to 0.
List the equilibria in order on a vertical line.

↓

2. SIGN CHART
In each region between equilibria, pick one y. Find the sign of $\boxed{f}(y)$.

↓

3. PHASE LINE
Draw an arrow in each region: $+$ up, $-$ down.

↓

4. CLASSIFY
Look at the arrows on both sides of each equilibrium. Use the table on the left.

↓

5. LIMIT (NO SOLVING)
Find the region of y_0. Follow its arrow to the nearest equilibrium in that direction.
That is $\lim_{t \to \infty} y(t)$.
No equilibrium in that direction: $\pm\infty$. y_0 on an equilibrium: it stays.

Lesson 9 — applications

THE IDEA
Every application is the same chain: words → a DE → separable or linear → solve → fit the constants. Usually two constants: C from the IC, and a rate constant k from a second data point.

The recipe

EXPLANATION

Words → math

Words	Math
"rate of change of Q"	$\frac{dQ}{dt}$
"proportional to"	$k\cdot$
"at a rate equal to 25% of"	$0.25\cdot$
"decreases", "decays", "loses"	a minus sign
"initially"	$t=0$
"6 months ago"	pick $t=0$ as now or as then. Stay consistent
"of the remaining"	of the current amount Q

Trap: Write the units of t down once. Every rate uses them.

WORK

1. TRANSLATE
Words → a DE. Use the table on the left.

↓

2. CLASSIFY
Separable (L4) or linear (L3)?

↓

3. SOLVE
General solution with C (and k).

↓

4. C FROM THE IC
Plug in $t=0$.

↓

5. k FROM A SECOND DATA POINT
Plug in (t_1, value_1). Solve for k: usually a ln.

↓

6. ANSWER THE QUESTION
A value at a time: plug in t. A time: solve for t with ln.

Newton's law of cooling

EXPLANATION

Pieces

Piece	Where to find it
T_m	the surrounding (room) temperature
T_0	the temperature at $t=0$
k	the rate constant, $k>0$. Unknown
(t_1, T_1)	the second reading

Trap: Subtract T_m first. The ratio is $\frac{T_1-T_m}{T_0-T_m}$, not $\frac{T_1}{T_0}$.
Trap: $T \to T_m$ as $t \to \infty$, from above or below.

WORK

1. DE
$\frac{dT}{dt} = -\boxed{k}\left(T - \boxed{T_m}\right)$ *(separable and linear)*

↓

2. SHAPE
$T = \boxed{T_m} + \left(\boxed{T_0} - \boxed{T_m}\right)e^{-\boxed{k}t}$

↓

3. k FROM THE READING
$e^{-\boxed{k}\boxed{t_1}} = \frac{\boxed{T_1}-\boxed{T_m}}{\boxed{T_0}-\boxed{T_m}} \Rightarrow \boxed{k} = -\frac{1}{\boxed{t_1}}\ln(\text{that ratio})$

↓

4. ANSWER
T at a time: plug in t. Time to reach T: $t = -\frac{1}{k}\ln\frac{T-T_m}{T_0-T_m}$.

Exponential growth and decay

EXPLANATION

Pieces

Piece	Where to find it
M_0	the amount at $t=0$
k	the rate. $k>0$ grows, $k<0$ decays
data	a second amount (t_1, M_1), or a half-life T

Which k?

Words	k
"decays at 25% of the remaining mass per hour"	$k=-0.25$ directly
"loses 25% each hour" (the amount after 1 h)	$e^k = 0.75$, $k = \ln 0.75$
amount M_1 at t_1	$k = \frac{\ln(M_1/M_0)}{t_1}$
half-life T	$\frac{1}{2} = e^{kT}$, $k = -\frac{\ln 2}{T}$

WORK

1. DE
$\frac{dM}{dt} = \boxed{k}M$ *(separable)*

↓

2. SHAPE
$M = \boxed{M_0}e^{\boxed{k}t}$

↓

3. k FROM THE DATA
Use the "Which k?" table on the left.

↓

4. ANSWER
M at a time: plug in t. Time to reach M: $t = \frac{\ln(M/M_0)}{k}$.

Mixing (tank)

EXPLANATION

Pieces

Piece	Where to find it
$Q(t)$	the amount of salt. The unknown
c_{in}	the concentration coming in. Pure water: 0
r_{in}, r_{out}	the flow in and the flow out (volume per time)
V_0, Q_0	the starting volume and amount
Δ	$r_{\text{in}} - r_{\text{out}}$, the change in volume per time

Lesson 3 pieces

Piece	Value
$p(t)$	$\dfrac{r_{\text{out}}}{V_0 + \Delta t}$
$q(t)$	$c_{\text{in}}\, r_{\text{in}}$
μ, $\Delta = 0$	$e^{(r_{\text{out}}/V_0)t}$
μ, $\Delta \neq 0$	$(V_0 + \Delta t)^{r_{\text{out}}/\Delta}$ *(a power of $V(t)$)*

Trap: Rate out uses the tank concentration $\frac{Q}{V}$, not c_{in}.
Trap: Concentration $= \frac{Q}{V}$. If asked for concentration, divide at the end.
Trap: Units: (amount/volume)(volume/time) = amount/time.

WORK

1. VOLUME
$V(t) = \boxed{V_0} + (\boxed{r_{\text{in}}} - \boxed{r_{\text{out}}})t = \boxed{V_0} + \boxed{\Delta}t$

2. DE: IN − OUT
$$\frac{dQ}{dt} = \boxed{c_{\text{in}}}\,\boxed{r_{\text{in}}} - \frac{Q}{V(t)}\,\boxed{r_{\text{out}}}$$

3. STANDARD FORM (LINE 1)
$Q' + \dfrac{\boxed{r_{\text{out}}}}{V(t)}\,Q = \boxed{c_{\text{in}}}\,\boxed{r_{\text{in}}}$ Read p, q (left table).

4. BUILD AND CHECK μ (LINES 2–3)
$\mu = e^{\int p\,dt}$. Use the left table. Check $\mu' = p\mu$.

5. MULTIPLY, COLLAPSE (LINES 4–5)
$(\mu Q)' = \mu\,\boxed{c_{\text{in}}}\,\boxed{r_{\text{in}}}$

6. INTEGRATE (LINE 6)
$$\mu Q = \int \mu\,\boxed{c_{\text{in}}}\,\boxed{r_{\text{in}}}\,dt + C$$

7. DIVIDE (LINE 7)
$$Q = \frac{1}{\mu}\left(\int \mu\, c_{\text{in}} r_{\text{in}}\,dt + C\right)$$

8. C FROM Q_0
Plug in $t = 0$, $Q = \boxed{Q_0}$.

Growth proportional to a power

EXPLANATION

Pieces

Piece	Where to find it
a	the power: "proportional to $\sqrt{W}$" gives $a = \frac{1}{2}$
k	the unknown constant
two data points	$(0, W_0)$ gives C, (t_1, W_1) gives k

Trap: $\int W^{-1/2}\,dW = 2W^{1/2}$. Divide by $\frac{1}{2}$.
Trap: Find C first from $t = 0$. Then k.

WORK

1. DE
$$\frac{dW}{dt} = \boxed{k}\,W^{\boxed{a}} \quad \textit{(separable)}$$

2. SEPARATE
$W^{-\boxed{a}}\,dW = \boxed{k}\,dt$

3. INTEGRATE
$$\frac{W^{1-\boxed{a}}}{1-\boxed{a}} = \boxed{k}t + C \quad (a = \tfrac{1}{2}\text{: } 2\sqrt{W} = kt + C)$$

4. C, THEN k
Plug in $(0, W_0)$ for C. Then (t_1, W_1) for k.

5. FREE W
$a = \frac{1}{2}$: $W = \left(\dfrac{kt + C}{2}\right)^2$

Lesson 10 — second order, constant coefficients

EXPLANATION

THE IDEA

Guess $y = e^{rx}$. Each derivative only multiplies by r, so the DE becomes a quadratic in r. Its two roots give the two pieces y_1, y_2. The type of roots picks the shape of the answer.

Pieces

Piece	Where to find it
a	the number on y'', with all terms on the left
b	the number on y'. Missing term: $b = 0$
c	the number on y. Missing term: $c = 0$
r_1, r_2	you make them: the roots of $ar^2 + br + c = 0$
α, β	complex roots only: $r = \alpha \pm \beta i$, $\beta > 0$
x_0, k_0, k_1	the ICs $y(x_0) = k_0$, $y'(x_0) = k_1$. Use them only at the end

The three cases ($D = b^2 - 4ac$)

Roots	General solution
$D > 0$: two real, $r_1 \neq r_2$	$c_1 e^{r_1 x} + c_2 e^{r_2 x}$
$D = 0$: one repeated, $r_1 = r_2 = r$	$c_1 e^{rx} + c_2 x e^{rx}$
$D < 0$: complex, $\alpha \pm \beta i$	$e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$
special: $\pm\beta i$ ($\alpha = 0$)	$c_1 \cos \beta x + c_2 \sin \beta x$
special: $r = 0$ is a root	$e^{0x} = 1$: that piece is a plain constant

What is happening in each case

Case	Why this shape
$r_1 \neq r_2$	two different exponentials. Neither is a multiple of the other, so they are a full set
$r_1 = r_2$	the quadratic gives only one exponential. xe^{rx} also solves the DE, and it is not a multiple of e^{rx}
$\alpha \pm \beta i$	$e^{(\alpha \pm \beta i)x} = e^{\alpha x}(\cos \beta x \pm i \sin \beta x)$. Add and subtract the two: the i drops out. α sets growth or decay, β sets the frequency

Read the pieces

Quick example	Work → answer
$y'' - 5y' + 6y = 0$	$(r-2)(r-3)$: $2, 3$. $c_1 e^{2x} + c_2 e^{3x}$
$y'' + 4y' + 4y = 0$	$(r+2)^2$: -2 twice. $(c_1 + c_2 x)e^{-2x}$
$y'' + 2y' + 5y = 0$	$D = -16$: $-1 \pm 2i$. $e^{-x}(c_1 \cos 2x + c_2 \sin 2x)$
$y'' + 9y = 0$	$b = 0$: $r^2 = -9$, $\pm 3i$. $c_1 \cos 3x + c_2 \sin 3x$
$y'' - 4y = 0$	$r^2 = 4$, ± 2. $c_1 e^{2x} + c_2 e^{-2x}$
$y'' + 3y' = 0$	$c = 0$: $r(r+3)$, $0, -3$. $c_1 + c_2 e^{-3x}$

Trap: $r^2 + 9 = 0$ gives $\pm 3i$ (cos, sin). $r^2 - 9 = 0$ gives ± 3 (exponentials). Watch the sign.
Trap: Repeated root: without the x you have the same solution twice.
Trap: β is the positive real number next to i. No i in the answer.

WORK

Steps

1. STANDARD FORM, READ THE PIECES

$\boxed{a}y'' + \boxed{b}y' + \boxed{c}y = 0$ *(the right side must be 0)*

↓

2. CHARACTERISTIC EQUATION

$\boxed{a}r^2 + \boxed{b}r + \boxed{c} = 0$

($y = e^{rx}$: $y' = re^{rx}$, $y'' = r^2 e^{rx}$. Divide by $e^{rx} \neq 0$. So $y'' \to r^2$, $y' \to r$, $y \to 1$)

↓

3. ROOTS

Factor, or $r = \dfrac{-\boxed{b} \pm \sqrt{D}}{2\boxed{a}}$, $D = \boxed{b}^2 - 4\boxed{a}\boxed{c}$

↓

4. PICK THE CASE

$D > 0$, $D = 0$, or $D < 0$: use the table on the left.

Complex: $\boxed{\alpha} = -\dfrac{b}{2a}$, $\boxed{\beta} = \dfrac{\sqrt{4ac - b^2}}{2a}$

↓

5. WRITE THE GENERAL SOLUTION

$D > 0$: $y = c_1 e^{\boxed{r_1}x} + c_2 e^{\boxed{r_2}x}$

$D = 0$: $y = c_1 e^{\boxed{r}x} + c_2 x e^{\boxed{r}x}$

$D < 0$: $y = e^{\boxed{\alpha}x}(c_1 \cos \boxed{\beta}x + c_2 \sin \boxed{\beta}x)$

↓

6. DIFFERENTIATE (IF AN IC IS GIVEN)

Find y' of the whole answer. Use the table below.

↓

7. IC

$y(\boxed{x_0}) = \boxed{k_0}$ and $y'(\boxed{x_0}) = \boxed{k_1}$: two equations. Solve for c_1, c_2.

Step 6 — derivatives

Piece	Derivative
e^{rx}	re^{rx}
xe^{rx}	$(1 + rx)\,e^{rx}$ *(product rule)*
$e^{\alpha x}\cos \beta x$	$e^{\alpha x}(\alpha \cos \beta x - \beta \sin \beta x)$
$e^{\alpha x}\sin \beta x$	$e^{\alpha x}(\alpha \sin \beta x + \beta \cos \beta x)$

Step 7 — the two equations when $x_0 = 0$

Case	$y(0) = k_0$ **and** $y'(0) = k_1$
$r_1 \neq r_2$	$c_1 + c_2 = k_0$, $r_1 c_1 + r_2 c_2 = k_1$
$r_1 = r_2 = r$	$c_1 = k_0$, $rc_1 + c_2 = k_1$
$\alpha \pm \beta i$	$c_1 = k_0$, $\alpha c_1 + \beta c_2 = k_1$

Trap: Differentiate first. Then plug in x_0.

Lesson 11 — Euler (Cauchy–Euler) equations

EXPLANATION

THE IDEA

In an Euler equation the power of x equals the order of the derivative. Guess $y = x^r$: every term becomes a number times x^r, so the DE again becomes a quadratic in r. The three cases are the Lesson 10 cases with $e^{rx} \to x^r$ and $x \to \ln x$.

Pieces

Piece	Where to find it
a	the number on x^2y''
b	the number on xy'. Missing term: $b = 0$
c	the number on y. Missing term: $c = 0$
r_1, r_2, α, β	you make them: the roots of $ar^2 + (b-a)r + c = 0$
x_0, k_0, k_1	the ICs. $x_0 \neq 0$, often $x_0 = 1$

Euler or not?

Equation	Verdict
$x^2y'' + 3xy' + y = 0$	Euler: x^2 on y'', x on y', number on y
$2x^2y'' - xy' = 0$	Euler, $c = 0$
$x^2y'' + y = 0$	Euler, $b = 0$
$y'' + xy' + y = 0$	not Euler: the powers do not match the orders

The three cases

Roots of $ar^2 + (b-a)r + c$	General solution ($x > 0$)
two real, $r_1 \neq r_2$	$c_1x^{r_1} + c_2x^{r_2}$
one repeated, r	$x^r(c_1 + c_2\ln x)$
complex, $\alpha \pm \beta i$	$x^\alpha(c_1\cos(\beta\ln x) + c_2\sin(\beta\ln x))$

What is happening: Lesson 10 → Lesson 11

Lesson 10	Lesson 11
e^{rx}	x^r
the extra x (repeated root)	the extra $\ln x$
$e^{\alpha x}$	x^α
$\cos\beta x, \quad \sin\beta x$	$\cos(\beta\ln x), \quad \sin(\beta\ln x)$

(Put $x = e^t$ ($t = \ln x$). The Euler equation becomes $ay_{tt} + (b-a)y_t + cy = 0$: a Lesson 10 equation in t. Its answer, with $t \to \ln x$, is the table. $x^{\beta i} = e^{i\beta\ln x} = \cos(\beta\ln x) + i\sin(\beta\ln x)$.)

Read the pieces

Quick example	Work → answer
$x^2y'' - 2y = 0$	$r^2 - r - 2$: $2, -1$. $\quad c_1x^2 + c_2x^{-1}$
$x^2y'' - 3xy' + 4y = 0$	$r^2 - 4r + 4$: 2 twice. $\quad x^2(c_1 + c_2\ln x)$
$x^2y'' + xy' + 4y = 0$	$r^2 + 4$: $\pm 2i$. $\quad c_1\cos(2\ln x) + c_2\sin(2\ln x)$
$x^2y'' + 3xy' + 5y = 0$	$r^2 + 2r + 5$: $-1 \pm 2i$. $\quad x^{-1}(c_1\cos(2\ln x) + c_2\sin(2\ln x))$

Trap: The indicial equation is $ar^2 + (b-a)r + c$, not $ar^2 + br + c$. The $r(r-1)$ gives the $-a$.
Trap: $b = 0$ still gives $-ar$: $\quad x^2y'' - 2y = 0 \to r^2 - r - 2$.
Trap: $x = 0$ is always a bad x: $p = \frac{b}{ax}$. The answer lives on $x > 0$. For $x < 0$, write $|x|$.

WORK

Steps

1. MATCH, READ THE PIECES

$\boxed{a}x^2y'' + \boxed{b}xy' + \boxed{c}y = 0$

(power of x = order of the derivative)

2. GUESS $y = x^r$

$y' = rx^{r-1}, \quad y'' = r(r-1)x^{r-2}$

($x^2 \cdot x^{r-2} = x^r$ and $x \cdot x^{r-1} = x^r$: every term has x^r)

3. INDICIAL EQUATION

$\boxed{a}r(r-1) + \boxed{b}r + \boxed{c} = 0$

Expand: $\boxed{a}r^2 + (\boxed{b} - \boxed{a})r + \boxed{c} = 0$

4. ROOTS, PICK THE CASE

Same test as Lesson 10, on the expanded equation: $D = (b-a)^2 - 4ac$.

Complex: $\boxed{\alpha} = -\dfrac{b-a}{2a}, \quad \boxed{\beta} = \dfrac{\sqrt{-D}}{2a}$

5. WRITE THE GENERAL SOLUTION

$D > 0$: $\quad y = c_1x^{\boxed{r_1}} + c_2x^{\boxed{r_2}}$

$D = 0$: $\quad y = x^{\boxed{r}}(c_1 + c_2\ln x)$

$D < 0$: $\quad y = x^{\boxed{\alpha}}(c_1\cos(\boxed{\beta}\ln x) + c_2\sin(\boxed{\beta}\ln x))$

6. DIFFERENTIATE (IF AN IC IS GIVEN)

Find y' of the whole answer. Use the table below.

7. IC

$y(\boxed{x_0}) = \boxed{k_0}, \quad y'(\boxed{x_0}) = \boxed{k_1}$. Solve for c_1, c_2. *($x_0 \neq 0$)*

Step 6 — derivatives

Piece	Derivative
x^r	rx^{r-1}
$x^r\ln x$	$x^{r-1}(r\ln x + 1)$
$\cos(\beta\ln x)$	$-\dfrac{\beta}{x}\sin(\beta\ln x)$ *(chain rule)*
$\sin(\beta\ln x)$	$\dfrac{\beta}{x}\cos(\beta\ln x)$

Step 7 — the two equations when $x_0 = 1$

$\ln 1 = 0$ and $1^r = 1$, so the equations are the Lesson 10 equations at $x_0 = 0$:

Case	$y(1) = k_0$ **and** $y'(1) = k_1$
$r_1 \neq r_2$	$c_1 + c_2 = k_0, \quad r_1c_1 + r_2c_2 = k_1$
$r_1 = r_2 = r$	$c_1 = k_0, \quad rc_1 + c_2 = k_1$
$\alpha \pm \beta i$	$c_1 = k_0, \quad \alpha c_1 + \beta c_2 = k_1$

Lesson 12 — nonhomogeneous: $y = y_c + y_p$

THE IDEA

A nonhomogeneous equation $y'' + py' + qy = f$ with $f \neq 0$ always has an answer in two parts: $y = y_c + y_p$. y_c solves the equation with f replaced by 0 (Lesson 10 or 11). It holds both constants. y_p is any one solution of the full equation. It holds no constants.

The shape of the general solution

EXPLANATION

Pieces

Piece	Where to find it
$p(x)$, $q(x)$	the coefficients of y' and y, after you divide by the coefficient of y''
$f(x)$	the right side (the forcing), after you divide. Not 0
$y_c = c_1 y_1 + c_2 y_2$	complementary: solve the partner with $f \to 0$
y_p	particular: one solution of the full equation. Given, or guess and check

Names

Name	Means
y_c or y_h	complementary = homogeneous part. Two constants
y_p	particular. No constants
$\{y_1, y_2\}$	fundamental set: two independent solutions of the $f \to 0$ partner
general solution	$y = y_c + y_p$: every solution, for some c_1, c_2

Read the pieces

Quick example	Work → answer
$y'' - 3y' + 2y = 4$	$c_1 e^x + c_2 e^{2x}$. Try 2: $0 - 0 + 4$ ✓ $y = c_1 e^x + c_2 e^{2x} + 2$
$y'' + y = x$	$c_1 \cos x + c_2 \sin x$. Try x: $0 + x$ ✓
$y'' - y = e^{2x}$	$c_1 e^x + c_2 e^{-x}$. Try Ae^{2x}: $4A - A = 1$, $A = \frac{1}{3}$
$x^2 y'' - 2y = -2$	$c_1 x^2 + c_2 x^{-1}$ (L11). Try 1: $0 - 2 = -2$ ✓

Trap: The ICs go on the full $y_c + y_p$, never on y_c alone. Add y_p first, then solve for c_1, c_2.
Trap: y_p has no constants. All the constants live in y_c.
Trap: To get the partner, change only the right side to 0. The left side stays the same.

WORK

Steps

1. STANDARD FORM, READ THE PIECES
$y'' + \boxed{p}y' + \boxed{q}y = \boxed{f}$ *(coefficient of y'' is 1)*

↓

2. HOMOGENEOUS PARTNER
Replace f with 0. Solve it: Lesson 10 (numbers) or Lesson 11 (Euler).
$\boxed{y_c} = c_1 y_1 + c_2 y_2$

↓

3. CHECK THE FUNDAMENTAL SET
$W = y_1 y_2' - y_1' y_2 \neq 0$ *(next page. Lesson 10 and 11 pairs always pass)*

↓

4. ONE PARTICULAR SOLUTION
Get $\boxed{y_p}$: given, or guess a shape like f and solve for its number.
Verify: put y_p in the left side. You must get f.

↓

5. ASSEMBLE
$y = \boxed{c_1 y_1 + c_2 y_2} + \boxed{y_p}$

↓

6. DIFFERENTIATE THE WHOLE y
$y' = c_1 y_1' + c_2 y_2' + \boxed{y_p'}$ *(y_p' too)*

↓

7. IC
$y(x_0) = k_0$, $y'(x_0) = k_1$: two equations. Solve for c_1, c_2.

Guess a shape for y_p (simple f)

If f is	try $y_p =$
a number k	A
e^{kx}	Ae^{kx}
a polynomial of degree n	a full polynomial of degree n
$\cos kx$ or $\sin kx$	$A\cos kx + B\sin kx$

Trap: If the guess is already part of y_c, it gives 0 on the left. Multiply the guess by x.

(Full method: Lesson 13. An f like $\tan x$ or $\frac{1}{x}$: Lesson 15.)

Properties

EXPLANATION

The operator

Write $L[y] = y'' + p\,y' + q\,y$. The homogeneous equation is $L[y] = 0$. The nonhomogeneous one is $L[y] = f$.

Properties

Property	Says
1. linear	$L[c_1u + c_2v] = c_1L[u] + c_2L[v]$. Every property below comes from this
2. superposition ($f = 0$)	$L[y_1] = 0$, $L[y_2] = 0 \Rightarrow L[c_1y_1 + c_2y_2] = 0$. That is why y_c is a combination
3. difference	$L[u] = f$, $L[v] = f \Rightarrow L[u - v] = 0$. Two particular solutions differ by a piece of y_c. So any one y_p works
4. superposition (f)	$L[y_{p1}] = f_1$, $L[y_{p2}] = f_2 \Rightarrow L[y_{p1} + y_{p2}] = f_1 + f_2$. Split f into parts, find one y_p for each part, add
5. scaling	$L[y_p] = f \Rightarrow L[ky_p] = kf$
6. not closed	two solutions of $L[y] = f$ add to a solution of $L[y] = 2f$, not of $L[y] = f$
7. zero	$y = 0$ solves $L[y] = 0$, never $L[y] = f$
8. nothing is lost	every solution of $L[y] = f$ is $y_c + y_p$ for some c_1, c_2. No lost solutions (compare L4)
9. count	order 2: two constants, two ICs (y and y' at the same x_0)

Read the pieces — superposition

Quick example	Work → answer
$y'' + y = x + 4$	$L[x] = x$, $L[4] = 4$: $y_p = x + 4$
$y'' - y = 3e^{2x}$	$L[\frac{1}{3}e^{2x}] = e^{2x}$, so $y_p = 3 \cdot \frac{1}{3}e^{2x} = e^{2x}$
$y'' - 3y' + 2y = 4$	2 and $2 + 5e^x$ both work: they differ by $5e^x$, a piece of y_c

WORK

Existence and uniqueness (Lesson 5, second order)

If p, q, f are continuous on an open interval I that contains x_0, then $y(x_0) = k_0$, $y'(x_0) = k_1$ has exactly one solution, and it lives on all of I.

Quick example	Work → answer
$y'' + \frac{1}{x-3}y = \ln x$, $x_0 = 1$	bad: $x \le 0$, 3. $(0, 3)$
$x^2y'' - 2y = 0$, $x_0 = 2$	$\div x^2$: bad 0. $(0, \infty)$

Trap: Use the Lesson 5 steps with p, q and f. Divide by the coefficient of y'' first.

The Wronskian: is $\{y_1, y_2\}$ a fundamental set?

1. BOTH SOLVE THE PARTNER
Check $L[y_1] = 0$ and $L[y_2] = 0$. *(Lesson 1 verify)*

↓

2. WRONSKIAN
$W = \boxed{y_1}\boxed{y_2'} - \boxed{y_1'}\boxed{y_2}$

↓

3. VERDICT
$W \neq 0$ at one x in I: fundamental set, $c_1y_1 + c_2y_2$ is the general solution.
$W = 0$ everywhere: one is a multiple of the other. Not a general solution.

(Abel: on I, W is either never 0 or always 0. Check at one easy x.)

Wronskians of the Lesson 10 and 11 pairs

y_1, y_2	W
e^{r_1x}, e^{r_2x}	$(r_2 - r_1)\,e^{(r_1+r_2)x}$
e^{rx}, xe^{rx}	e^{2rx}
$e^{\alpha x}\cos\beta x$, $e^{\alpha x}\sin\beta x$	$\beta\,e^{2\alpha x}$
x^{r_1}, x^{r_2}	$(r_2 - r_1)\,x^{r_1+r_2-1}$
x^r, $x^r \ln x$	x^{2r-1}
$x^\alpha\cos(\beta\ln x)$, $x^\alpha\sin(\beta\ln x)$	$\beta\,x^{2\alpha-1}$

(Every one is $\neq 0$: the case recipes always give a fundamental set.)

Trap: e^{rx}, e^{rx} gives $W = 0$. That is why a repeated root needs the extra x (or $\ln x$).

Lesson 13 — undetermined coefficients

THE IDEA
The coefficients are numbers, and f is built from polynomials, e^{kx}, cos and sin. Every derivative of such an f stays in the same family. So guess a y_p with the same shape as f and unknown numbers $A, B, \ldots$. Substitute, match like terms, and solve for the numbers. If the plain guess is already part of y_c, multiply it by x.

Build the guess

EXPLANATION

Pieces

Piece	Where to find it
a, b, c	the numbers on y'', y', y. Numbers only
$f(x)$	the right side
k or $k \pm \beta i$	the **f-number**: read it off f (step 3)
n	the highest power of x in f. No x: $n = 0$
r_1, r_2	the roots of $ar^2 + br + c$, from y_c (Lesson 10)
s	how many of r_1, r_2 equal the f-number: 0, 1 or 2
$A, B, \ldots$	the unknown numbers in y_p

The guess table (then multiply by x^s)

If f is	**f-number → try** $y_p = x^s \cdot (\ldots)$
a number	$0 \to A$
x^2 (degree 2)	$0 \to Ax^2 + Bx + C$
e^{3x}	$3 \to Ae^{3x}$
xe^{-x}	$-1 \to (Ax + B)e^{-x}$
$\cos 2x$ or $\sin 2x$	$\pm 2i \to A\cos 2x + B\sin 2x$
$e^x \sin 3x$	$1 \pm 3i \to e^x(A\cos 3x + B\sin 3x)$
$x\cos x$	$\pm i \to (Ax + B)\cos x + (Cx + D)\sin x$

Read the pieces

Quick example	Work → answer
$y'' - 3y' + 2y = 4x$	$1, 2$; 0: $s = 0$. $Ax + B$
$y'' - y' - 2y = e^{2x}$	$2, -1$; 2 once: $s = 1$. Axe^{2x}
$y'' + 4y' + 4y = e^{-2x}$	-2 twice; -2: $s = 2$. Ax^2e^{-2x}
$y'' + y = \sin x$	$\pm i$; $\pm i$: $s = 1$. $x(A\cos x + B\sin x)$
$y'' + y = e^x$	$\pm i$; 1: $s = 0$. Ae^x
$y'' + 3y' = 6$	$0, -3$; 0: $s = 1$. Ax

Trap: Trig: keep both cos and sin in the guess, even when f has only one.
Trap: Polynomial: keep every lower power. $f = x^2$ needs $Ax^2 + Bx + C$.
Trap: x^s multiplies the whole guess, every term.
Trap: Compare the whole f-number. e^{-2x} with a root 2: no match. $\sin x$ with roots $\pm 2i$: no match.

WORK

Steps

1. STANDARD FORM, READ THE PIECES
$\boxed{a}y'' + \boxed{b}y' + \boxed{c}y = \boxed{f}$
(numbers only. Not numbers, or a hard f: Lesson 15)

↓

2. y_c (LESSON 10)
Roots r_1, r_2 of $\boxed{a}r^2 + \boxed{b}r + \boxed{c} = 0$. $y_c = c_1y_1 + c_2y_2$

↓

3. READ f
$\boxed{f} =$ (degree $\boxed{n}$ polynomial) $\cdot\, e^{\boxed{k}x} \cdot$ (cos $\boxed{\beta}x$ or sin $\boxed{\beta}x$)
f-number: k. With trig: $k \pm \beta i$.
(a missing part: no polynomial, $n = 0$; no e, $k = 0$)

↓

4. COUNT THE MATCHES
$s =$ how many of r_1, r_2 equal the f-number: 0, 1, or 2.
(a match means the plain guess is part of y_c: the left side gives 0)

↓

5. WRITE THE GUESS
$\boxed{y_p} = x^s \cdot$ (the table on the left)
Full polynomials of degree n. Both cos and sin.

f is a sum

SPLIT
$f = f_1 + f_2$: do steps 3–5 for each part, each with its own s. Solve each part. Then $y_p = y_{p1} + y_{p2}$.
(Lesson 12, property 4)

Quick example	Work → answer
$y'' - y = x + e^x$	± 1. x: 0, $s = 0$, $A + Bx$. e^x: 1, $s = 1$, Cxe^x

Trap: Do not use one guess for the whole sum. Each part has its own f-number, so its own s.

The family test

If f is	then
x^n, e^{kx}, cos, sin, and products	Lesson 13
$\tan x$, $\sec x$, $\ln x$, $\frac{1}{x}$, $\frac{e^x}{x}$	not in the family: Lesson 15

($\tan x$ has derivatives $\sec^2 x$, $2\sec^2 x\tan x$, $\ldots$: a new shape every time, so no finite guess.)

Solve for the numbers

EXPLANATION

Step 6 — derivatives of the common guesses

y_p	y_p' and y_p''
$Ax+B$	A and 0
Ae^{kx}	kAe^{kx} and k^2Ae^{kx}
Axe^{kx}	$A(1+kx)e^{kx}$ and $A(2k+k^2x)e^{kx}$
Ax^2e^{kx}	$A(2x+kx^2)e^{kx}$ and $A(2+4kx+k^2x^2)e^{kx}$
$A\cos\beta x+B\sin\beta x$	$\beta(-A\sin\beta x+B\cos\beta x)$ and $-\beta^2(A\cos\beta x+B\sin\beta x)$
$x(A\cos\beta x+B\sin\beta x)$	$(A\cos\beta x+B\sin\beta x)+\beta x(-A\sin\beta x+B\cos\beta x)$ and $2\beta(-A\sin\beta x+B\cos\beta x)-\beta^2x(A\cos\beta x+B\sin\beta x)$

Shortcut: $f=e^{kx}$

Write $P(r)=ar^2+br+c$, the characteristic polynomial.

Case	y_p
$s=0$: $P(k)\neq 0$	$\dfrac{e^{kx}}{P(k)}$
$s=1$: k is a simple root	$\dfrac{xe^{kx}}{P'(k)}=\dfrac{xe^{kx}}{2ak+b}$
$s=2$: k is a double root	$\dfrac{x^2e^{kx}}{2a}$

(For $f=Me^{kx}$, multiply by M (L12, property 5). Use it to check step 9.)

Read the pieces — match

Quick example	Work → answer
$y''-3y'+2y=4x$, $Ax+B$	$-3A+2Ax+2B=4x$. x: $2A=4$. 1: $-3A+2B=0$. $y_p=2x+3$
$y''-y=3\cos 2x$, $A\cos 2x+B\sin 2x$	$-5A\cos 2x-5B\sin 2x=3\cos 2x$. $A=-\frac{3}{5}$, $B=0$
$y''+y=\sin x$, $x(A\cos x+B\sin x)$	$-2A\sin x+2B\cos x=\sin x$. $A=-\frac{1}{2}$, $B=0$. $y_p=-\frac{1}{2}x\cos x$
$y''-y'-2y=e^{2x}$, Axe^{2x}	$3Ae^{2x}=e^{2x}$. $A=\frac{1}{3}$ *(shortcut: $2(2)-1=3$)*

Trap: A term missing from f has coefficient 0. $f=\sin x$: the $\cos x$ equation is $\ldots=0$.
Trap: With e^{kx} or trig and $s\geq 1$, every term with the extra x cancels. If one is left, recheck s or a derivative.

WORK

Steps (continued)

6. DIFFERENTIATE THE GUESS
$\boxed{y_p'}$ and $\boxed{y_p''}$. Use the table on the left.

↓

7. SUBSTITUTE
$\boxed{a}\boxed{y_p''}+\boxed{b}\boxed{y_p'}+\boxed{c}\boxed{y_p}=\boxed{f}$

↓

8. COLLECT, MATCH
Group the left side by like terms: each power x^j, each e^{kx}, each cos, each sin. Coefficient on the left = coefficient in f (0 if missing). One equation per term.

↓

9. SOLVE FOR THE NUMBERS
Solve for $A, B, \ldots$. Put them back into y_p.
(polynomials: start at the highest power. Each equation gives one number)

↓

10. ASSEMBLE
$y=\boxed{c_1y_1+c_2y_2}+\boxed{y_p}$

↓

11. IC (IF GIVEN)
Differentiate the whole y. Plug in x_0. Solve for c_1, c_2.
(Lesson 12, steps 6–7)

Why the extra x

If the guess is part of y_c, then $L[\text{guess}]=0$: it can never equal f. Multiplying by x gives a new function that is not in y_c. A double root needs x^2, as in Lesson 10.
Trap: Find y_p first. The ICs go on y_c+y_p, never on y_c alone.

Lesson 14 — reduction of order

EXPLANATION

THE IDEA

You know one solution y_1. Guess $y = u\,y_1$ with an unknown function u. The plain u term drops out, because y_1 already solves the equation. Only u' and u'' are left: a first order equation in $w = u'$. Solve for w, integrate once for u, and $y_2 = u\,y_1$.

Pieces

Piece	Where to find it
$p(x)$	the coefficient of y', after you divide by the coefficient of y''. Sign included
y_1	given. It solves the $f \to 0$ partner
$u(x)$	the unknown multiplier. You make it
$w = u'$	the new unknown. First order
$f(x)$	the right side, after you divide. Usually 0

Shortcut ($f = 0$)

Formula	In words
$w = \dfrac{e^{-\int p\,dx}}{{y_1}^2}$	steps 3–5, done once in general
$y_2 = y_1 \displaystyle\int \frac{e^{-\int p\,dx}}{{y_1}^2}\,dx$	the whole method in one line

Read the pieces

Quick example	Work → answer
$y'' - 4y' + 4y = 0$, e^{2x}	$p = -4$: $w = \frac{e^{4x}}{e^{4x}} = 1$, $u = x$. $y_2 = xe^{2x}$
$x^2y'' - 3xy' + 4y = 0$, x^2	$p = -\frac{3}{x}$: $w = \frac{x^3}{x^4} = \frac{1}{x}$, $u = \ln x$. $y_2 = x^2 \ln x$
$x^2y'' + xy' - y = 0$, x	$p = \frac{1}{x}$: $w = \frac{x^{-1}}{x^2} = x^{-3}$, $u = -\frac{1}{2x^2}$. $y_2 = \frac{1}{x}$
$xy'' - (x+1)y' + y = 0$, e^x	$p = -1 - \frac{1}{x}$: $w = \frac{xe^x}{e^{2x}} = xe^{-x}$, $u = -(x+1)e^{-x}$. $y_2 = x+1$

Equivalences

This	is the same as
the u term in step 3	$u \cdot L[y_1] = u \cdot 0$: it always drops
${y_1}^2\,w$	the Wronskian $W(y_1, y_2)$. $e^{-\int p\,dx}$ is Abel's formula
the extra x (L10) and $\ln x$ (L11)	what this method gives: rows 1 and 2 above

Trap: Divide by the coefficient of y'' first. p keeps its sign, and $e^{-\int p}$ adds a second minus.

Trap: A plain u term left in step 3 means y_1 is not a solution, or a derivative is wrong.

Trap: No $+C$ in w or u (when $f = 0$). You need one y_2. Drop constant factors: $-\frac{1}{2x} \to \frac{1}{x}$.

WORK

Steps

1. STANDARD FORM, READ THE PIECES

$y'' + \boxed{p}\,y' + q\,y = \boxed{f}$

Check that y_1 solves it with $f \to 0$. *(Lesson 1 verify)*

↓

2. GUESS $y = uy_1$

$y = \boxed{u}\,\boxed{y_1}$

$y' = u'y_1 + uy_1'$

$y'' = u''y_1 + 2u'y_1' + uy_1''$ *(product rule twice)*

↓

3. SUBSTITUTE — THE u TERM DROPS

$\boxed{y_1}\,u'' + \left(2\boxed{y_1'} + \boxed{p}\,\boxed{y_1}\right) u' = \boxed{f}$

(the u terms are $u(y_1'' + py_1' + qy_1) = u \cdot 0$)

↓

4. REDUCE: $w = u'$

$\boxed{y_1}\,\boxed{w'} + \left(2\boxed{y_1'} + \boxed{p}\,\boxed{y_1}\right)\boxed{w} = \boxed{f}$

(order 2 → order 1: the name of the method)

↓

5. SOLVE FOR w ($f = 0$: SEPARABLE, L4)

$$\frac{\boxed{w}'}{\boxed{w}} = -\frac{2\boxed{y_1'}}{\boxed{y_1}} - \boxed{p} \quad \Rightarrow \quad \boxed{w} = \frac{e^{-\int \boxed{p}\,dx}}{\boxed{y_1}^2}$$

($\int \frac{2y_1'}{y_1}\,dx = 2\ln|y_1|$, so $e^{-2\ln y_1} = \frac{1}{y_1^2}$)

↓

6. INTEGRATE

$\boxed{u} = \displaystyle\int \boxed{w}\,dx$ *(no $+C$)*

↓

7. y_2, GENERAL SOLUTION

$y_2 = \boxed{u}\,\boxed{y_1}$. Drop constant factors.

$y = c_1\boxed{y_1} + c_2\,y_2$ *(check $W = y_1^2 w \neq 0$)*

Step 5 when $f \neq 0$ (linear, L3)

LINE 1 — DIVIDE BY y_1

$$\boxed{w}' + \left(\frac{2\boxed{y_1'}}{\boxed{y_1}} + \boxed{p}\right)\boxed{w} = \frac{\boxed{f}}{\boxed{y_1}}$$

↓

LINES 2–7 — SEVEN LINES ON w

$\mu = \boxed{y_1}^2\,e^{\int \boxed{p}\,dx}$. Then lines 3–7 of Lesson 3.

↓

6–7. INTEGRATE, KEEP BOTH $+C$

$\boxed{u} = \int \boxed{w}\,dx + C_2$. $y = \boxed{u}\,\boxed{y_1}$ is then the whole general solution.

Trap: With $f \neq 0$, keep the $+C$ in line 6 and in u. Those two constants are c_1, c_2.

Lesson 15 — variation of parameters

EXPLANATION

THE IDEA

Take $y_c = c_1 y_1 + c_2 y_2$ and let the constants vary: $y_p = u_1 y_1 + u_2 y_2$. Two unknown functions need two conditions. You choose one, and the DE gives the other. The 2×2 system gives u_1' and u_2' directly. It works for any f and any p, q, once you have y_1, y_2.

Pieces

Piece	Where to find it
$a(x)$	the coefficient of y''. Divide by it first
$f(x)$	the right side, after you divide by a
y_1, y_2	a fundamental set: Lesson 10, 11, 14, or given
W	you make it: $y_1 y_2' - y_1' y_2$
u_1, u_2	the unknown functions. You make them

Lesson 13 or Lesson 15?

If	Use
the coefficients are not numbers	L15
f is $\tan x$, $\sec x$, $\ln x$, $\frac{1}{x}$, $\frac{e^x}{x}$, …	L15: no finite guess
numbers, and f is in the L13 family	L13: algebra, no integrals
both work	L13 is usually faster. L15 always works

Where the formulas come from

Condition	Says
1. you choose	$u_1' y_1 + u_2' y_2 = 0$. Then $y_p' = u_1 y_1' + u_2 y_2'$: no u'' can appear
2. the DE forces	$u_1' y_1' + u_2' y_2' = f$ *(put y_p in; the $L[y_1]$, $L[y_2]$ terms are 0)*
Cramer's rule	the system has determinant W: $u_1' = \dfrac{-y_2 f}{W}$, $u_2' = \dfrac{y_1 f}{W}$

Read the pieces

Quick example	Work → answer
$y'' + y = \sec x$	$\cos x$, $\sin x$, $W = 1$. $u_1' = -\tan x$, $u_2' = 1$. $y_p = \cos x \ln\lvert\cos x\rvert + x \sin x$
$y'' - 2y' + y = \dfrac{e^x}{x}$	e^x, xe^x, $W = e^{2x}$. $u_1' = -1$, $u_2' = \frac{1}{x}$. $-xe^x + xe^x \ln\lvert x\rvert$. Drop $-xe^x$: $y_p = xe^x \ln\lvert x\rvert$
$x^2 y'' - 2xy' + 2y = x^3$	$f = x$, not x^3. x, x^2, $W = x^2$. $u_1' = -x$, $u_2' = 1$. $y_p = -\frac{x^3}{2} + x^3 = \frac{x^3}{2}$

Trap: Divide by a first. $x^2 y'' - \ldots = x^3$ has $f = x$, not x^3. This is the most common error.
Trap: The minus sign is on u_1'. Each u' uses the other y: u_1' has y_2, u_2' has y_1.
Trap: No $+C$ in u_1, u_2. The constants only rebuild y_c.

WORK

Steps

1. STANDARD FORM
$y'' + py' + qy = \boxed{f}$
(divide every term by a, the right side too)

2. FUNDAMENTAL SET
$y_c = c_1 \boxed{y_1} + c_2 \boxed{y_2}$ *(Lesson 10, 11, 14, or given)*

3. WRONSKIAN
$\boxed{W} = \boxed{y_1}\boxed{y_2'} - \boxed{y_1'}\boxed{y_2}$ *(table in Lesson 12. Never 0)*

4. THE TWO DERIVATIVES
$\boxed{u_1'} = -\dfrac{\boxed{y_2}\boxed{f}}{\boxed{W}}$ $\boxed{u_2'} = \dfrac{\boxed{y_1}\boxed{f}}{\boxed{W}}$
Simplify each one before you integrate.

5. INTEGRATE
$\boxed{u_1} = \int \boxed{u_1'}\,dx$ $\boxed{u_2} = \int \boxed{u_2'}\,dx$ *(no $+C$)*

6. BUILD y_p
$\boxed{y_p} = \boxed{u_1}\boxed{y_1} + \boxed{u_2}\boxed{y_2}$
Simplify. A piece that is a multiple of y_1 or y_2 can be dropped: it is part of y_c.

7. ASSEMBLE, IC
$y = c_1 \boxed{y_1} + c_2 \boxed{y_2} + \boxed{y_p}$
ICs on the whole y. *(Lesson 12, steps 6–7)*

If you forget the formulas

SOLVE THE SYSTEM
$u_1' y_1 + u_2' y_2 = 0$
$u_1' y_1' + u_2' y_2' = f$
Eliminate one unknown, or use Cramer's rule.

In one line: $y_p = -y_1 \displaystyle\int \frac{y_2 f}{W}\,dx + y_2 \int \frac{y_1 f}{W}\,dx$

Step 5 — integrals

Integrand	Antiderivative
$\tan x$	$-\ln\lvert\cos x\rvert = \ln\lvert\sec x\rvert$
$\sec x$	$\ln\lvert\sec x + \tan x\rvert$
$\sec x \tan x$ / $\sec^2 x$	$\sec x$ / $\tan x$
$\sin^2 x$ / $\cos^2 x$	$\frac{x}{2} - \frac{\sin 2x}{4}$ / $\frac{x}{2} + \frac{\sin 2x}{4}$
$\frac{1}{x}$	$\ln\lvert x\rvert$
xe^{kx}	$\frac{1}{k^2}(kx-1)\,e^{kx}$ *(by parts)*
$\ln x$	$x \ln x - x$ *(by parts)*